DOMAIN 3 RISK MANAGEMENT

656 When designing a safety program, it is essential that psychological factors like individual differences, motivation, emotion, attitudes, and _____ be considered.

1) Reward
2) Punishment
3) Learning
4) Negativism

Ans.3 Learning underlies much of what makes for differences and similarities among people. Through learning, people have developed certain kinds of psychological needs, habitual patterns of behavior, ways of reacting to emotion, and the attitudes which brought them into the workplace. It is essential to consider learning because training is a major consideration in safety programming.

657 Non-regulatory consequences of failure to adequately clean-up chemical releases include all EXCEPT:

1) Decreased property values
2) Difficulties in selling the affected property
3) Reduced collateral value of the property in borrowing situations
4) Reduced charges to waste disposal budget

Ans.4 Reduced charges to waste disposal budget is not possible.(Most likely, costs will increase.) Chemical contamination can cause a property to have a negative economic value because the costs of remediation exceed the market value of the property, once remediated. There are legal/regulatory consequences as well.

658 The total value estimated to be at risk as a result of a particular hazard being evaluated is called:

1) Loss exposure
2) Total risk exposure
3) Risk unit
4) Loss expected

Ans.2 Total risk exposure is the total number of dollars estimated to be at risk as a result of a particular hazard being evaluated.

659 In assessing risk, one of the analytical techniques used by systems safety is called hazard risk. It is:

1) A means used to identify and measure the hazard
2) A way to measure potential hazard conditions in the operational phase
3) A method for calculating the effect of a hazard if allowed to become an accident
4) The product of hazard severity and hazard possibility of occurrence

Ans.4 Risk assessment techniques use mainly system-oriented inductive and deductive analyses. The system safety analytical technique called hazard risk is the product of hazard severity and hazard possibility of occurrence (frequency).

660 System availability is defined as:

1) A measure of the degree to which an item is in an operable and committable state.
2) The hardware which maintains the higher state of productivity
3) The average degree of reliability needed maintain to status quo
4) Simply stated by the formula: As = (An)(Aq)(Ai)

Ans.1 System availability is defined as a measure of the degree to which an item is in an operable and committable state.

661When using a FTA for accident investigation, the accident is the top event. What is a significant reason for using this technique to analyze the accident?

1) All the events are identified
2) The probability that the accident can occur again can be calculated
3) It speeds up the investigation
4) Events that have a zero probability of occurrence are eliminated and the remainder are investigated

Ans.4 Often it is important to eliminate those items that did not contribute to an accident so they do not become the focus of investigation and change.

662 Statistically, when several events are independent they:

1) Are also mutually exclusive
2) Do not affect one another
3) Are not capable of happening at the same time
4) Are also conditional on one another

Ans.2 Independent events do not have an effect on the probability of occurrence of any other event. Mutually exclusive events cannot happen together. It is possible that independent events could happen at the same time.

663Two events that cannot occur together are called:

1) Independent events
2) Non-random events
3) Mutually exclusive events
4) Bayseian events

Ans.3 Mutually exclusive means that the events cannot occur at the same time. Sometimes this is confused with independent events where one event does not influence another event.

664 Which of the following is not an element of a risk assessment?

1) Hazard identification
2) Exposure assessment
3) Risk communication
4) Dose-response assessment

Ans.3 Risk communication is the process of informing the public in a understandable and effective manner of the potential harm from a given exposure situation and/or particular agent. Hazard identification, dose-response assessment, and exposure assessment are all elements of risk assessment as outlined by the National Academy of Sciences.

665 The most appropriate method of analysis to completely evaluate a product in a safety program is:

1) Failure mode and effects analysis
2) Fault-tree analysis
3) Preliminary hazard analysis
4) No single method of analysis

Ans.4 While all are appropriate techniques for use in evaluation of a product, no single method can be used to completely evaluate a product.

666 When using system safety techniques to evaluate a product, the most appropriate time to conduct the first analysis is:

1) When the patent is applied for
2) When the product is first conceived
3) When the product prototype is developed
4) When the first products are manufactured but before it is distributed

Ans.2 While system safety techniques may be employed throughout the manufacturing process, the first analysis should be done at the time the product is conceived. This helps assure that safety is built in from the start and may help avoid costly decisions to re-engineer the product or perhaps even drop plans for the manufacture and distribution of the product.

667 Systems safety techniques employ the use of four categories of analysis. Fault tree analysis is ALWAYS a member of which category:

1) Deductive analyses
2) Inductive analyses
3) Quantitative hazard analyses
4) Qualitative hazard analyses

Ans.1A fault tree analysis is an example of a deductive analysis. Deductive analyses start with a top-level event, and logically determine its specific causes. Some FTA's can be qualitative - some can be quantitative. No FTA is inductive.

668 The technique that provides a means for quantitatively evaluating the contribution of human error to the degradation of product quality is called a:

1) Fault hazard analysis
2) Failure modes effects analysis
3) Fault tree analysis
4) THERP

Ans.4THERP, a technique for human error predictions was developed by the Scandia Corporation. It provides a means for quantitatively evaluating the contributions of human error to the degradation of product quality. It can be used for human components in systems and can be combined with either failure modes effects or fault tree analysis.

669 In system safety you would use which Boolean gate to connect the causative agents "exothermic reaction" and "outside heat source" to the end product "heat".

1) "AND"
2) "OR"
3) Either
4) Neither

Ans.2 You would utilize an "OR" gate because either an "exothermic reaction" OR an "outside source" will cause heat.

670 In performing a quantitative fault tree analysis, the first step is to define the top event to be investigated by the tree. Then you must gain an understanding of the system. Which of the following is NOT one of the remaining steps?

1) Construct the tree
2) Collect qualitative data
3) Evaluate the probability of the event chosen as the event to be investigated
4) Analyze computer output

Ans.2 Quantitative data collection is the fourth step, not qualitative data collection.

671 The first step of a system-level fault tree analysis is to define the top event. The failure selected to define the event must be:

1) Major system failures
2) Procedure failures
3) Component failures
4) All failures

Ans.1 Major system failures are chosen because they will determine key improvements that must be made in system safety. Selection of other, lower-level failures would result in overlooking much of the system for potential failures.

672 The value of λ in a fault tree analysis indicates:

1) An event expected to happen during normal use.
2) The hazard rate
3) The failure rate
4) The error rate

Ans.3 The Greek letter "⌊" represents the probability of a failure of a component of a part of a system. It can also indicate the mean time between failures.

673 The method that uses mathematical simulation, chance processes, and random numbers to determine whether and how frequently the outcome of an event or complex relationship will occur is known as:

1) Mathematical Simulation
2) Monte Carlo Method
3) Modeling Method
4) Computer Simulation

Ans.2 The Monte Carlo Method of simulation is the name of the method used to analyze complex trees. It requires the use of a computer since a reasonably complex tree will require significant calculations to get a reliable answer for the reliability of the main event.

674 The "bathtub" curve is a stress related depiction that describes the relationship between component failure and exposure. Failures occurring during early stages of use occur in the debugging period, which is also called:

1) Experimental period
2) Burn-in period
3) Burn-out period
4) Assessment period

Ans.2 Failures occurring during early stages of use occur during the burn-in period.

675If the mean time between failures of a system is 5,000 trials, what is the reliability of the system over 15,000 trials?

1) 0.05
2) 0.23
3) 0.33
4) 3.00

Ans.1 Using the reliability formula: $R_t = e^{-\lambda\tau}$
λ = 1/mean time between failures
$R_t = e^{(-15,000/5,000)}$
$R_t = 0.0498$

676If the reliability of a system over 15,000 trials is 0.05, what is the probability of failure of the system?

1) 67%
2) 76%
3) 95%
4) 100%

Ans.3 Using the formula: P(f) = 1 - R(t), where:
P(f) is the probability of failure and R(t) is the reliability over time.
P(f) = 1 - 0.05
= 0.95 or 95%

677The reliability of a component that has two parallel redundant circuits (each redundant system has a reliability of 0.85) is:

1) 1.00
2) 0.98
3) 0.89
4) 0.85

Ans.2 The probability of failure of each leg is calculated as R = (1 - 0.85) = 0.15
The probability of a system failure is:
Pf = 0.15 * 0.15 = 0.0225
The system reliability is Rs = 1 - 0.0225 = 0.9775

678A system that has multiple units that must operate in order for a given output to occur is said to have:

1) Parallel redundancy
2) Fail safe characteristics
3) High failure rates
4) Series redundancy

Ans.4 Such a system has series redundancy, and as such, has a low probability for a failure to occur.

679A system with two components in parallel, each with a reliability of 0.70, would have a reliability of:

1) 0.30
2) 0.50
3) 0.70
4) 0.91

Ans.4 $R_{series} = R_1 R_2$
$R_{parallel} = 1 - (U_1 U_2)$
$U_1 = 1 - 0.70 = 0.30$
$U_2 = 1 - 0.70 = 0.30$
$R_{parallel} = 1 - (0.30)(0.30)$
$= 1 - 0.09$
$= 0.91$
If one component fails, another can act as a backup.

680If a system consists of 100 components in series, each with a reliability of 0.9900, the reliability of the entire system is:

1) 0.366
2) 0.634
3) 0.990
4) 0.999

Ans.1 In series, multiply the reliabilities of all components.
$R = 0.99^{100}$
$= 0.366$
366 times out of 1,000 the system would be able to properly perform its function.

681For reliability purposes, adding components in parallel:

1) Decreases reliability in the system
2) Has no effect on reliability in the system
3) Increases the reliability of the system
4) Depends on other subsystems

Ans.3 Rseries = R_1R_2
Rparallel = 1 - (U_1 U_2)
$U_1 = 1 - R_1$
$U_2 = 1 - R_2$
Assume $R_1 = 0.9$ and $R_2 = 0.8$
In a series system the system reliability is:
Rs = 0.9 * 0.8 = 0.72
In a parallel system the system reliability is:
Rp = - (0.1 * 0.2) = 0.98

682For system (A) to fail, components (B) and (C) which are in parallel, must fail. If the reliability of (B) and (C) are both 0.70, then what is the reliability of system (A)?

1) 0.49
2) 0.51
3) 0.91
4) 0.97

Ans.3 In a parallel arrangement the reliability of system
RA = 1 - (1 - 0.70) (1 - 0.70)
= 0.91

683The reliability of a system is enhanced by adding a component in:

1) Parallel
2) Series
3) Sequence
4) Progression

Ans.1 The reliability is enhanced by adding a component in parallel. The additional component, which performs the same function within a given system, makes it redundant (or parallel).

684 A switch has 5 components. Each part has a failure rate of 0.3. Failure of any part causes the switch to fail. What in the reliability of the switch system?

1) 1.50
2) 0.168
3) 0.150
4) 0.00243

Ans.2 Reliability is defined as (1 - failure rate). The reliability of this system is the product of the reliabilities of the components.
$R = (1 - 0.3)^5 = 0.168$

685A switch has 5 components, all redundant to one another. The component failure rates are 1×10^{-3}, 1×10^{-2}, 1×10^{-1}, 1×10^{-3}, and 1×10^{-2}, respectively. What is the system failure rate?

1) 1×10^{-11}
2) 5×10^{-11}
3) 1.21×10^{-1}
4) 1.21×10^{-2}

Ans.1 Given the parallel construction, the total failure rate is multiplicative.
Failure Rate = $(1 \times 10^{-3})(1 \times 10^{-2})(1 \times 10^{-1})(1 \times 10^{-3})(1 \times 10^{-2}) = 1 \times 10^{-11}$

686You are asked to recommend a parallel or series design for a critical switch. The switch has two components with a failure rate of 1×10^{-5} and 1×10^{-8}. Which should you choose?

1) Parallel construction
2) Series construction
3) Double parallel construction
4) Double series construction

Ans.1 To reduce the failure rate you would choose the parallel construction.
The parallel failure rate is:
Failure Rate = $(1 \times 10^{-5})(1 \times 10^{-8}) = 1 \times 10^{-13}$
The series construction failure rate is:
Failure Rate (s) = $(1 \times 10^{-5}) + (1 \times 10^{-8}) = 1 \times 10^{-5}$

687When assessing the reliability of a system, all of the following are considered appropriate EXCEPT:

1) Part usage
2) Functional requirements
3) Environmental requirements
4) Reliability bonding

Ans.4 Reliability bonding is not considered appropriate when assessing the reliability of a system.

688In a large sample of components, 5% were found to be defective. If 7 components are sampled, what is the probability that exactly 3 will be defective?

1) 0.999
2) 0.036
3) 0.001
4) 0.0001

Ans.4$P(d) = Q^y [P(s)^{(n-y)}]$
y = # of successes
n = # of trials
Q = Probability of Failure
P(s) = Probability of Success
P(d) = Probability of Defective
$P(d) = (0.05)^3[0.95^{(7-3)}]$
$= 0.000125 * 0.95)^4$
$= 0.000125 * 0.857$
$= 0.000107$

689When performing a hypothesis test, a Type I error is:

1) Accepting a null hypothesis when it is true
2) Rejecting a null hypothesis when it is true
3) Accepting a null hypothesis when it is false
4) Rejecting a null hypothesis when it is false

Ans.2 Statistically it is possible to get a sample where the sample statistic falls outside the acceptance region but the null hypothesis is indeed true. This is called a Type I error.

690Accepting the null hypothesis when it is false is called a:

1) Type I Error
2) Type II Error
3) Type III Error
4) Type IV Error

Ans.2 Accepting the null hypothesis when it is false is called a Type II error and is symbolized by the Greek letter beta.

691When testing an assumption about a population mean and we accept the alternative hypothesis:

1) The sample mean must have fallen within the acceptance b region
2) The sample mean must have fallen within the rejection region
3) The population mean must have fallen within the acceptance region
4) The population mean must have fallen within the rejection region

Ans.2 If the sample mean falls within the rejection region, we reject the null hypothesis and accept the alternative hypothesis.

692 In a behavioral observation process, the principle method of feedback is usually considered to be:

1) Posting group data on percent of safe behaviors observed
2) Discipline when an employee is seen doing something incorrectly
3) Posting individual employee results
4) Supervisors are informed of their group's performance

Ans.1 In a behavioral observation process, feedback is generally given by posting group results such as the percent of safe behaviors observed.
E. Scott Geller, The Psychology of Safety

693The psychologist considered to be the "father" of behavior modification is:

1) Sigmund Freud
2) B. F. Skinner
3) Douglas McGregor
4) Frederick Herzberg

Ans.2 B. F. Skinner is generally considered to be the father of behavioralism.
Dan Petersen, Safety Management: A Human Approach

694Which of the following statements is NOT true concerning human behavior?

1) There is no such person as an average person
2) Best approach is to design systems to fit percentiles of the population
3) In fitting personnel to a job consider both the physical and personality characteristics
4) Attitude is an important characteristicof fitting the worker to the task

Ans.4 Attitude is not an important characteristic of fitting the worker to the task.

695Which of the following is NOT true concerning critical incident technique when evaluating human behavior?

1) Take a structured sample of employees
2) Interview to collect accident information on near misses, difficulties in operations, and conditions that could have resulted in death or property loss
3) Incidents are classed into hazard categories and problem areas or behaviors are identified
4) Periodically repeat the procedure as conditions or populations change

Ans.1 Taking a RANDOM sample of employees is a critical incident technique when evaluating human behavior.

696Which one of the following is NOT true concerning behavior sampling when evaluating human behavior?

1) Use a behavior or activity sampling sheet
2) Do walk-around inspections and record the number and types of behavior observed
3) Make the observations at the same time each day
4) Calculate the percentage of times certain behaviors are observed

Ans.3 Making observations at DIFFERENT times of the day, days of week, and shifts is true when using behavior sampling.

697"Risk analysis" is normally conducted before

1) Dose response modeling
2) Environmental analysis
3) Fault tree analysis
4) Risk management

Ans.4 The methodology of risk analysis is quite different from risk management. Risk analysis is a scientific activity, the result of which is an estimate of risk. Risk management must determine whether the risk is acceptable and if not, what methods will be used to reduce the risk to an acceptable level.

698If the chi-square statistic of a sample corresponds to a p of 0.01, what may one infer?

1) The result will occur 1% of the time
2) The mean of the sample will occur 1% of the time
3) The result is statistically significant at the p = 0.01 level
4) The result is most likely a matter of chance

Ans.3A chi-square statistic equal to 0.01 means that the result is statistically significant with a probability of error of 1% (0.01).

699An inductive technique that explores different responses to "challenges" is called:

1) Event tree analysis
2) Fault tree analysis
3) Logic analysis
4) Bottom-up assessment

Ans.1 An event tree analysis is an inductive technique that explores different responses to "challenges".

700 This technique envisions a "primitive" or unprotected system and systematically evaluates the effects of adding various controls through a brainstorming approach.

1) Brainstorming
2) Primitive analysis
3) Naked man
4) Hazard control review

Ans.3 Naked man is the technique that envisions a "primitive" or unprotected system and systematically evaluates the effect of adding various controls through a brainstorming approach.

701A technique that utilizes a method of developing job procedures that includes a systematic task analysis that examines each step, and identifies preventive or corrective actions required to ensure a safe operation is called a(n):

1) Job safety analysis
2) Task analysis
3) Operability study
4) Systematic task analysis

Ans.1A job safety analysis is a technique that utilizes a method of developing job procedures that includes a systematic task analysis that examines each step, and identifies preventive or corrective actions required to ensure a safe operation.

702This technique utilizes a six-step process to analyze and control human errors. The process involves selecting an event, identifying the tasks associated with the event, separating specific behaviors of each task and assigning basic error rates. This is called a(n):

1) Behavioral assessment
2) Job safety analysis (JSA)
3) Technique for human error rate prediction (THERP)
4) Operability study

Ans.3"Technique for human error rate prediction", (THERP) utilizes a six-step process to analyze and control human errors.

703A technique that provides formal documentation and feedback to a safety analysis performed on changes to the end product throughout the life cycle is called a:

1) Life cycle analysis
2) Change analysis
3) Project evaluation tree
4) PERT

Ans.2A change analysis is a technique that provides formal documentation and feedback of safety analyses performed on changes (to the end product) throughout the life cycle.

704The combined couplet of frequency and severity is defined as:

1) Accident severity
2) Risk
3) Incident rate
4) Frequency rate

Ans.2 Risk is universally defined as the combination of frequency and severity, or consequence.

705The uncertainty of risk has which of the following components?

1) Likelihood and consequence
2) Consequence and exposure
3) Exposure
4) Exposure, consequence, and likelihood

Ans.4 Exposure, consequence, and likelihood contribute to the uncertainty of risk.

706 Three categories of hazards in hazard analysis studies are:

1) Errors of omission, errors of commission, and inherent properties
2) People, machines, and the environment
3) Environmental issues, human and material failure, and inherent properties
4) Material or human failure, environmental failure, and properties

Ans.3 The three broad categories are: environmental issues that create stress, inherent properties that create hazards, and failures of people and materials.

707 A FMEA (failure modes and effects analysis) is extremely effective when applied to:

1) Multiple failures of a system
2) Multiple failures of a product
3) Single failures of a unit
4) Analyze non-critical events
5)

Ans.3A FMEA is the right systems safety tool to use to analyze a single failure or a single unit failure. Often FMEA is used in conjunction with a fault tree analysis to evaluate a product's safety. Fault tree logic can identify possible failures and a FMEA can be used to analyze each failure event.

708 Which of the following is NOT a formal method for assessing risk?

1) Maintenance Mode Analysis
2) Preliminary Hazard Analysis
3) Failure Mode and Effects Analysis
4) Fault Tree Analysis

Ans.1 Many hazards are still possible in spite of product design efforts. They may result from component failures, malfunctions, operational methods, environments encountered, maintenance, and operation errors. These hazards can be identified by one or more of the analyses listed, all of which serve to accomplish a common function identified as risk assessment. The differences between the listed analyses is in that they occur at different times during a system safety program. A maintenance mode analysis is not a risk assessment method.

709 The technique in which the failure or malfunction of each component is considered and the effects of the failure traced upward through the system is identified with:

1) Failure modes and effects analysis
2) Preliminary hazard analysis
3) Fault tree analysis
4) THERP

Ans.1A failure modes and effects analysis is an analytical technique in which the failure or malfunction of each component is considered, including the mode of failures. The effects of the failure are traced upward through the entire system and its effect on the system performance is evaluated.

710 A system safety analysis can clarify a complex process by:

1) Making a step-by-step evaluation of each component that makes up the system
2) Developing a conceptual model of the system
3) Employing the use of applicable statistics
4) Both 1 and 3

Ans.2 An analysis of the system can clarify a complex process by devising a chart or model that provides a comprehensive, overall view of the process by showing its principal elements and the ways in which they are interrelated.

711 An event tree analysis is best defined as:

1) A method for identifying various possible outcomes
2) A system of numerical analysis
3) A method utilizing true and false options
4) A system of statistical analysis

Ans.1 An event tree analysis is a logic method that breaks down the possible outcomes of a single event.

712The first step in an event tree analysis is termed the:

1) Failure event
2) Primal event
3) Logic inhibit
4) Initiating event

Ans.4 The initial event in any event tree is termed the "initiating event".

713The effects of the failure are traced through the system and the ultimate effect on task performance is evaluated. This process is called a:

1) Preliminary hazard analysis
2) Fault hazard analysis
3) Failure modes and effects analysis
4) Critical item

Ans.3 When the effects of the failure are traced through the system and the ultimate effect on task performance is evaluated, this process is called a failure modes and effects analysis.

714The mobility of a system, subsystem, component or part from performing its function within specified limits, under specified conditions for a specified duration is called (a):

1) Malfunction
2) Hazard
3) Risk
4) Failure

Ans.4 When a system, subsystem, component, or part departs from its intended design parameters, the departure from design is called a failure. For example, if an electric motor is designed to produce 100 horsepower (+/- 1 horsepower) with a 40 amp current draw (+/- 1 amp), then the electric motor has experienced a failure if it produces only 50 horsepower when drawing 50 amps.

715A process to identify hazards and recommend risk reduction alternatives in procedurally controlled activities during all phases of intended use is called a(n):

1) Preliminary hazard analysis
2) Risk assessment
3) Operating and support hazard analysis
4) Fault tree analysis

Ans.3 An operating and support hazard analysis is a process to identify hazards and recommend risk reduction alternatives in procedurally controlled activities during all phases of intended use.

716A single item of hardware, the failure of which would lead directly to loss of life, vehicle, or mission is called a:

1) Critical hardware item
2) Risk item
3) Single failure point
4) Failure event

Ans.3A single failure point is a single item of hardware, the failure of which would lead directly to loss of life, vehicle, or mission.

717Those computer software components whose errors can result in a hazard or loss of predictability or control of a system are called:

1) Software code critical
2) Single failure point
3) Unsafe error
4) Safety-critical computer software components

Ans.4 Safety-critical computer software components are those computer software components whose errors can result in a hazard or loss or predictability or control of a system.

718The formal analysis of a system and the interrelationships among its various parts to determine the real and potential hazards within the system, and to suggest ways to reduce and control those hazards is called a:

1) Fault hazard analysis
2) Subsystem hazard analysis
3) System hazard analysis
4) Failure modes and effects analysis

Ans.3A system hazard analysis is a formal analysis of a system and the interrelationships among its various parts to determine the real and potential hazards within the system, and to suggest ways to reduce and control those hazards.

719 An analysis technique that is used primarily to evaluate multiple failures that may be caused by a single event or causal factor common to or shared by multiple components is a:

1) Failure modes and effect analysis
2) Common cause failure analysis
3) Fault hazard analysis
4) Fault tree analysis

Ans.2A common cause failure analysis is an analysis technique that is used primarily to identify a single event or causal factor common to or shared by multiple components. This analysis is looking for failures that can cause multiple "independent" safeguards to fail . For example, a common cause failure of redundant cooling water pumps may be the loss of electricity to both pumps.

720 A technique to determine an unintended energy route, which can allow an undesired function to occur, prevent functions from occurring, or adversely affect the timing of functions is called a:

1) Sneak circuit analysis
2) Fault hazard analysis
3) Fault tree analysis
4) Energy route review

Ans.1A sneak circuit analysis is a technique to determine an unintended energy route that can allow an undesired function to occur, prevent functions from occurring, or adversely affect the timing of functions.

721 An example of an inductive analysis is a:

1) Fault tree analysis
2) Failure mode and effects analysis
3) Accident investigation
4) Qualitative hazard analysis

Ans.2 Inductive analyses begin with a component or function, and then attempt to determine possible outcomes of failures of the function or component. It is a predictive tool for the user and does not rely on a history of failures to determine the future failures.

722Many of the potential failures of a system can be determined before manufacturing of the product by using a technique known as a:

1) PHA
2) FMEA
3) JSA
4) FHA

Ans.1A preliminary hazard analysis is a technique used to conduct an initial hazard evaluation that can then be used to make informed decisions about the product's design and manufacture.

723 The system safety technique that starts with an undesired event and analyzes the way the undesired event occurs is:

1) SADT
2) Fault tree analysis
3) PERT
4) Job Safety Analysis

Ans.2A fault tree analysis starts with the undesired event - the top-level event.

724 The most commonly used systems safety analysis technique is:

1) Fault tree analysis
2) Failure mode and effect analysis
3) Preliminary hazard analysis
4) Fault hazard analysis

Ans.3 Preliminary hazard analysis is the most common tool used by the systems safety practitioner. It is used at the beginning stages of the development of a product.

725An example of an inductive analysis technique is the:

1) Quantitative fault tree analysis
2) Qualitative fault tree analysis
3) Failure Modes, Effects, and Criticality analysis
4) Top-down analysis

Ans.3FMECA is an example of an inductive analysis.

726 Which of the following is not an inductive analysis technique?

1) FTA
2) JSA
3) FMEA
4) FMECA

Ans.1 The fault tree analysis (FTA) is an example of a deductive analysis technique.

727In order to use Boolean logic to simplify the calculations needed to analyze a fault tree:

1) An entry can only be in one of two possible modes: on-off, will occur-will not occur, closed-not closed
2) Intermediate conditions can be expressed and analyzed
3) Events are dependent on each other
4) Cause-effect relationships are never used

Ans.1 Entries are either on or off and can have no intermediate condition for a fault tree to be analyzed using Boolean logic.

728 When determining the relative safety of a product using fault tree analysis, a quick way to evaluate this safety is to:

1) Count the number of "OR" gates
2) Subtract the number of "OR" gates from the number of "AND" gates
3) Count the number of "AND" gates
4) Add the total number of gates in the tree except for the "inhibit" gates

Ans.3"AND" gates require that all events must occur in order for the failure to occur therefore, the probability of failure is the product of all the individual input probabilities. This value is much smaller than from the same events going through an "OR" gate .

729 A bleeder valve on a furnace fails after 20,000 days of operation. If the valve operates once per day what has been its success rate?

1) 0.99995
2) 0.99999
3) 0.99900
4) 0.99000

Ans.1 Success Rate = 19,999 successes / 20,000 attempts
= 0.99995 success rate

730 A bleeder valve on a furnace fails after 20,000 days of operation. If the valve operates once per day, what is its failure rate?

1) 1×10^{-4}
2) 5×10^{-4}
3) 1×10^{-5}
4) 5×10^{-5}

Ans.4 Failure Rate = 1 failure / 20,000 attempts
= 0.00005
= 5×10^{-5}

731A door hinge fails 4 times out of 1,000,000 operations. What is the failure rate?

1) 4×10^{-7}
2) 4×10^{-9}
3) 4×10^{-6}
4) 4×10^{-5}

Ans.3 Failure Rate = 4 failures / 1,000,000 attempts
= 4×10^{-6}

732 A door hinge fails 4 times out of 1,000,000 operations. What is its success rate?

1) 99.9996×10^{-2}
2) 99.9999×10^{-2}
3) 99.9×10^{-3}
4) 99×10^{-2}

Ans.1 Success Rate = 999,996 successes / 1,000,000 attempts
= 0.999996
= 99.9996×10^{-2}

733 A switch has 4 components. Each has a failure rate of "a". The components are connected in series. What is the switch's failure rate?

1) (4a) + 1
2) a^4
3) 4a
4) a(a+4)

Ans.3 For n components in series, the overall failure rate is the sum of the single components' failure rates.

734 A method of identifying errors and unsafe conditions that contribute to both potential and actual injurious accidents within a given population by means of a stratified, random sample of participants and observers selected from within this population is called:

1) Job safety analysis
2) Procedural analysis
3) Critical incident technique
4) Operating and support hazard analysis

Ans.3 Critical incident technique is a method of identifying errors and unsafe conditions that contribute to both potential and actual injurious accidents within a given population by means of a stratified random sample of participants and observers selected from within this population.

735 The major cause of on-the-job fatalities in the United States is:

1) Employee slips, trips, and falls
2) Employee motor vehicle use
3) Logging
4) Machinery

Ans.2 The single largest cause of on-the-job fatalities is from motor vehicles. They account for about a third of all deaths.

736 The difference between a mock-up and a breadboard model is:

1) The mock up is operational but not the same size as a full scale model. The breadboard is also operational but does not resemble the product
2) A breadboard functions and a mock-up does not
3) There are no differences
4) One is used to simulate a failure and one to simulate a product design

Ans.2A mock-up looks like the product but does not function. A breadboard works but does not resemble the final product.

737 The degree of safety achieved in a system depends directly on:

1) Good design practices
2) Employing systems safety techniques
3) Management emphasis
4) Meeting set objectives

Ans.3 The degree of safety achieved in a system depends directly on management emphasis. Government and contractors will apply management emphasis to safety during the system acquisition process and throughout the life cycle of each system, making sure mishap risk is understood and risk reduction is always considered in the management review process.

738 The uncertainty factor approach (i.e., derivation of a risk reference dose or RFD) is used to derive guidelines for exposure to _____ agents.

1) Carcinogenic
2) Noncarcinogenic
3) Toxicologic
4) Pharmacologic

Ans.2 Use of the RFD is based on the presumed threshold nature of these effects and uses uncertainty factors to attempt to account for intraspecies and interspecies variation.

739Children are generally _____ susceptible to the effects of a number of heavy metals as compared to adults.

1) Less
2) More
3) Equally
4) Variably

Ans.2 Children have an immature blood-to-brain (BTB) barrier. Mature BTB's are effective at preventing lead and other heavy metals from reaching the brain and spinal areas.

740A type of toxic reaction common in acute exposure to solvents is:

1) Spontaneous aborted pregnancy
2) Narcosis
3) Anemia
4) Pneumonitis

Ans.2 Acute exposure to a variety of solvents produces disturbances including disorientation, euphoria, giddiness, confusion, and eventually unconsciousness, paralysis, convulsions, and death if the exposure is not discontinued.

741The primary target tissue in methanol toxicity is the:

1) Cornea
2) Cerebellum
3) Nasal epithelium
4) Retina

Ans.4 Methanol intoxication is characterized by initial mild inebriation followed by an asymptomatic period of 12 to 24 hours after which metabolic acidosis develops. Visual problems include eye pain, blurred vision, reduced visual fields, and blindness (due to damage to the retinal tissue).

742 The most common type of cancer produced by ionizing radiation in children under 10 years of age is:

1) Osteogenic sarcoma
2) Myeloid leukemia
3) Chronic lymphatic leukemia
4) Thyroid cancer

Ans.2 Myeloid and acute leukemias are the most readily induced of the radiation-caused cancers; however, the incidence of chronic lymphatic leukemia does not appear to be affected by radiation.

743Which of the following designs will NOT allow equipment to operate?

1) Fail-active
2) Fail-passive
3) Fail-safe
4) Fail-operational

Ans.2A fail-passive system shuts down the equipment.

744 Which system design is intrinsically safe?

1) A non-redundant system
2) Double parallel system
3) Cost effective system
4) Single parallel system

Ans.2 The only intrinsically safe system listed is a double parallel system.

745 In system safety design, which design philosophy is the closest to "perfect" safety?

1) Intrinsically safe
2) Fail-safe
3) Fail-passive
4) Fail-operational

Ans.1 The design philosophy which is closest to "perfect" safety is intrinsic safety. This design philosophy seeks to limit failures to an extremely small percentage. All of the others allow failure, but seek to engineer out the hazard a failure would bring.

746 This system safety design reduces the system to its lowest energy level given a failure:

1) Intrinsically safe
2) Fail-passive
3) Fail-active
4) Fail-operational

Ans.2 In fail-passive designs, the system is brought to its lowest energy level, given a component failure. The other options do not reduce the system energy level to its lowest potential.

747 Which of the following designs are NOT examples of fail-safe design philosophy?

1) Fail-passive
2) Fail-active
3) Fail-operational
4) All are fail-safe designs

Ans.4 Fail-safe designs include all of the above as design philosophies.

748 All of the following are examples of malfunctions which impact a system EXCEPT:

1) Structural
2) Mechanical
3) Power
4) Human

Ans.4 The potential malfunctions in any system include structural failure, mechanical malfunctions, power failures, and electrical malfunctions. The human component, though critical, is not considered to be a malfunction.

749A design that automatically reduces the system to its lowest energy level is considered:

1) Fail-active
2) Fail-ceased
3) Fail-passive
4) Fail-non-operational

Ans.2 Fail-passive is a design that automatically reduces the system to its lowest level.

750Epidemiological studies are either descriptive or experimental. Descriptive epidemiology helps in preventing disease by:

1) Observing outcomes among study groups not created experimentally
2) Observing outcomes among study groups created experimentally
3) Examining dose response relationships in terms of time, place, and exposed individuals
4) Both 1 and 3

Ans.4 Descriptive epidemiology is the observation of outcomes among study groups not created experimentally followed by an examination of dose response relationships. The key concept is that the epidemiologist observes existing groups of people and obtains observation data. No hypothesis is being tested as in an experimental study.

751In determining the probability of a cause/effect relationship between an illness and an agent in the workplace, the first consideration must be to:

1) Examine the interplay between the physical environment and biological effects
2) Examine dose relationships in terms of time and place
3) Establish that a disease condition does exist
4) Determine toxicity of the chemical agent

Ans.3 The first consideration in determining the probability of a cause/effect relationship between an illness and an agent in the workplace is to establish that a disease condition actually exists or that the particular manifestations of the disease are the result of exposure to a specific harmful agent.

752 Which factor will LEAST likely determine whether or not an individual exposed to a biohazard will actually contract the disease?

1) The number of organisms in the work environment
2) The exposed individual's blood type
3) The resistance of the workers to the organism(s)
4) The virulence of the organism(s)

Ans.2 The combination of the number of organisms in the environment, the virulence of these organisms, the resistance of the workers to the organisms, and the presence of other chemical or physical stresses in the environment ultimately determines whether or not a person exposed to a biohazard will actually contract the disease. Blood type is the least likely factor.

753 Anthropometric data are widely used in:

1) Epidemiology
2) Ergonomics/human factors engineering
3) Biomechanics
4) Both 2 and 3

Ans.4 Anthropometry is concerned with measurement of the human body. Included in these measurements are body dimensions, range of body members, and muscle strength. Also considered are dimensions taken statically and dynamically. Viewed in physical terms, the body is a complex system of levers and motors attached to a frame. Biomechanics deals with the study and function of the body in relation to dynamics of men-machine systems. Ergonomics/HFE professionals use anthropometric data to design and construct equipment, tools, etc., that work safely and efficiently with humans.

754 The three primary methods, or types, of epidemiology studies are retrospective studies, prospective studies, and:

1) Case control studies
2) Cross-sectional studies
3) Cohort studies
4) Incident studies

Ans.2 Cross-sectional studies are, by definition, conducted on a sample which is drawn from one point in time (e.g., the population currently working in a manufacturing plant). The association between an exposure and effect at one specific point in time is studied. Employees that are no longer working within the subject area are not considered in the analysis. Therefore, since this type of study is based on a sample of subjects from one point in time, its ability to identify cause-effect relationships is very limited. These studies are quick and inexpensive, however, and can provide hypotheses for later comprehensive studies.

755 The amount of make-up air volume in most cases is determined by:

1) The number of air changes required
2) The total volume of air removed
3) The required comfort conditions
4) Carbon dioxide concentration

Ans.2 In most cases, make-up air volume should equal the total volume of air removed from the building by exhaust ventilation systems, process systems, and combustion processes. Determination of actual volumes of air removed usually requires an inventory of exhaust locations.

756 Automatic sprinkler systems seldom fail to control fires, however, when a failure does occur the dominant cause is:

1) Foreign material obstructing the piping system
2) Freezing of water in the piping system
3) Improperly closed supply valve
4) Inadequate supply of water

Ans.3 Factory Mutual records cite improperly closed PIV's to be the dominant cause for automatic sprinkler system failure. This occurs most often because of maintenance related actions that necessitate system shutdown followed by failure to open the valve after work has been completed. Shutting off the supply valves prematurely during a fire is another leading cause for failure.

757 Which of the following portable fire extinguishers should not be used to extinguish a small fire in a confined space?

1) ABC dry chemical
2) Carbon dioxide
3) Water
4) BC dry chemical

Ans.2 The carbon dioxide will replace oxygen in a confined space and could lead to asphyxiation.

758 All of the following are steps in a chemical risk analysis EXCEPT:

1) Identification of risk
2) Dose-response
3) Exposure assessment
4) Personal protective equipment required

Ans.4 Chemical risk analyses classically involve risk identification, quantification, dose-response development, and exposure assessment.

759 A life-threatening disease epidemic has increased the normal annual death rate from 18 deaths per 1,000 people to 28 deaths per 1,000. This is:

1) A death by disease rate of 1.0%
2) A death by disease rate of 1.8%
3) An increased threat to the population of 2,800 per year
4) Both 2 and 3

Ans.1 The 10 deaths per 1,000 people increase in the death rate is attributed to the life-threatening disease epidemic. Therefore, the death by disease rate is 10 per 1,000 people, or 1%. The usual death rate includes deaths by everything EXCEPT the disease.

760 "The equipment operator failed to shut down the boiler when the high pressure alarm activated in the control room". During an accident investigation, this will end up being a(n):

1) Causal factor
2) System effect
3) Accepted risk
4) Recognized fault

Ans.1 Causal factors are key events in the chain leading to an accident. If a causal factor is eliminated, the accident would likely not have happened. Causal factors are always in the form of "subject - verb".

761 Descriptions of the potential accidents that could result from the hazard in terms of number of persons affected, environmental impacts, and effects on other subsystems or the total system are called:

1) Causal factors
2) Hazards
3) System effects
4) Events

Ans.3 A description of potential accidents that could result from the hazard in terms of number of persons affected, environmental impacts, and effects on other subsystems of the total system are system effects.

762 A widely recognized protocol utilized to prevent the transmission of the HIV virus is called:

1) Aids quarantine
2) Universal precautions
3) Aids awareness
4) HBV vaccination

Ans.2 The terminology "universal precautions" is a set of work practices adopted by health care and related workers to prevent the spread of the HIV and other pathogenic organisms. 29 CFR 1910.1030 (d)(1)

763 In the late 1980s and early 1990s, which vaccine was identified as important for personnel at risk of exposure to bloodborne pathogens?

1) Tetanus
2) HIV
3) HBV
4) HAV

Ans.3 The Hepatitis B vaccine was determined to be critical to offer to healthcare workers. In 1991 it was made mandatory for workers at risk of exposure to bloodborne pathogens in the United States. 29 CFR 1910.1030 (f)(1)

764 Human factors engineering or ergonomics is fitting the worker to a task or the task to the worker. In a man/machine system, the machine is better at all of the following EXCEPT:

1) Memory
2) Language ability
3) Exerting large forces smoothly
4) Rapid computations

Ans.2 Man is better at language ability.

765 Safety behavior modification programs have the following purpose:

1) Eliminate unsafe acts
2) Increase safe acts
3) Both 1 and 2
4) None of the above

Ans.3 Behavior modification programs generally try to eliminate unsafe acts and increase safe acts.
Dan Petersen, Safety Management: A Human Approach

766 The most important sensory input for human decision-making is:

1) Visual
2) Auditory
3) Tactile
4) Olfactory

Ans.1 Visual input is the most important. Eye-hand coordination is required for the successful completion of most workplace operations.

767 In behavioral terms, "ABC" refers to:

1) Activator, behavior, consequence
2) Attitude, behavior, consequence
3) A nice way of saying, "keep it simple stupid"
4) The ease with which people's behavior can be understood

Ans.1 ABC refers to the activator, behavior, consequence model of behavior.

E. Scott Geller, The Psychology of Safety

768 Behavior is most influenced by:

1) Activators
2) Consequences
3) Discipline
4) Feedback

Ans.2 While all four answers have an impact on behavior, consequences, in general, have the most impact. Discipline and feedback can be considered types of consequences, but there are others to be considered including positive reinforcement and reward.

E. Scott Geller, The Psychology of Safety.

769 Which of the following characteristics of consequences have the most influence on increasing desired behaviors?

1) Negative, future, uncertain
2) Negative, immediate, certain
3) Positive, future, certain
4) Positive, immediate, certain

Ans.4 Consequences which are positive, immediate, and certain have the most impact on behavior.
E. Scott Geller, The Psychology of Safety

770 Which one of the following is NOT a recommended engineering control method?

1) Isolation or enclosure
2) Substitution of less toxic materials
3) Use of NIOSH-approved respirators
4) Local exhaust ventilation

Ans.3 Use of respirators is not an engineering control.

771 In doing a risk assessment for your hazard analysis, what is the most appropriate definition of risk?

1) The product of hazard severity and hazard probability
2) Psychrometric hazard perception
3) Anthropologic perception of hazard
4) Hazard fault

Ans.1 In hazard analysis the correct definition of risk is the hazard severity times the hazard probability. Answers 2 and 3 are social risk parameters that have little value in system safety.

772 Reliability-centered maintenance (RCM) is an approach to equipment maintenance starting with a facility or system analysis to develop a cost-effective regime for:

1) Repairing/maintaining systems when they fail
2) Repairing/maintaining systems based on historical failure data
3) Repairing/maintaining systems based on the results of constant equipment monitoring data
4) Both 2 and 3

Ans.4 RCM is a high-level process for maintaining equipment that focuses on preventing problems through equipment monitoring (vibration tests, oil tests, thermography, etc.) as well as considering the criteria for equipment routine and preventive maintenance frequency (from equipment criticality data, etc.). RCM is designed for maintenance resources to be used in a highly structured way, focused on critical equipment and critical monitoring.

773 A system safety technique that starts with an undesired event and an analysis of how that event can occur is called?

1) Fault tree analysis
2) Boolean algebra
3) Preliminary hazard analysis
4) Both 1 and 3

Ans.1 Fault tree analysis starts with an undesired event (top-level event) and analyzes its antecedents.

774 In systems safety, logic gates are used in fault tree construction. An "AND" gate is used when:

1) Outputs occur if any inputs are present
2) Outputs occur if an input and a condition exist
3) Outputs occur if two or more necessary inputs are present
4) Outputs occur if all necessary inputs are simultaneously present

Ans.4 An "AND" gate is used if all inputs are present at the same time.

775 An "inhibit" gate is used in fault tree construction when:

1) An input produces an output all the time
2) An output inhibits another output
3) An output occurs if an input and a condition exist at the same time
4) An input does not always produce an output

Ans.3 An inhibit gate shows an input and an output along with the condition that must exist for the output to occur.

776 In systems safety fault tree analysis, a "random" gate is a variation of what type of gate?

1) AND
2) OR
3) NOT
4) Inhibit

Ans.4 A random gate is similar to an inhibit gate. In a random gate, it is drawn as a hexagon like an inhibit gate but an ellipse is attached to the side and shows the probability for the random event to occur.

777 Two valves are placed in parallel on a tank. Valve "A" has a failure rate of 1×10^{-3}. Valve "B" has a failure rate of 1×10^{-2}. The likelihood of both devices failing simultaneously is:

1) 1×10^{-2}
2) 1×10^{-3}
3) 1×10^{-5}
4) 1×10^{-10}

Ans.3 To determine the probability of this failure, two simultaneous events must occur, therefore, the probability of the two events is the product of the two probabilities.
Failure Rate = $(1 \times 10^{-3})(1 \times 10^{-2})$
= 1×10^{-5}

778 Valve system V_1 has the following valve component failure rates:
$X_1 = 0.4$
$X_2 = 0.1$
$X_3 = 0.2$
$X_4 = 0.5$
The fault tree of the valve system indicates that valve subsystem SS_1 contains series components X_1, X_2, and X_3. Valve component X_4 and SS_1 are parallel and redundant. What is the anticipated failure rate of V_1?

1) 0.35
2) 0.65
3) 0.75
4) 0.95

Ans.1 Add the series components for the failure rate of SS_1.
$?SS_1 = 0.4 + 0.1 + 0.2$
= 0.7
Multiply the failure rate of SS_1 with redundant valve component X_4 to calculate the failure rate of V_1.
$?V_1 = 0.5 \times 0.7$
= 0.35

779A switch is operated twice per day. If the switch fails after 10 years of operation, what is its failure rate?

1) 2.7×10^{-1}
2) 1.8×10^{-2}
3) 1.9×10^{-3}
4) 1.4×10^{-4}

Ans.4 Total Number of Ops = 10 years * 365 days/year * 2 ops/day
= 7,300 operations
Failure Rate = 1 failure/7,300 operations
= 0.00014
= 1.4×10^{-4}

780 Given a failure rate per year of 3×10^{-3} for a valve, if the valve operates once per year, how many failures will you expect in 1,000 years?

1) 0.03
2) 0.3
3) 3
4) 30

Ans.3 Multiply the failure rate per year by the number of years.
Failure Rate = 3×10^{-3} failures/year * 1,000 years
= 3 failures

781 A automotive parts distributor purchases parts from overseas locations and markets these parts in the United States. The distributor receives 25% of its parts from country "A" who has a 10% reject rate, 25% from country "B" who has a 5% reject rate, 12.5% from country "C" with a 15% reject rate, and 37.5% from country "D" with a 1% reject rate. If you purchase a defective part what is the probability it came from country "D"?

1) 42%
2) 31%
3) 21%
4) 6%
5)

Ans.4 Basis: 100,000 parts
Country "A" 25,000 * 10% = 2,500 rejects
Country "B" 25,000 * 5% = 1,250 rejects
Country "C" 12,500 * 15% = 1,875 rejects
Country "D" 37,500 * 1% = 375 rejects

P(A) = 2,500 / 6,000 = 0.4167
P(B) = 1,250 / 6,000 = 0.2083
P(C) = 1,875 / 6,000 = 0.3125
P(D) = 375 / 6,000 = 0.0625

782 The reliability department has determined that the MTBF (Mean Time Between Failures) for a computer disk drive is 95,000 hours. What is the probability that the disk drive will experience at least one failure in the first 10 years of operation (87,000 hours)?

1) 75%
2) 60%
3) 25%
4) 2%

Ans.2 $P_f = 1 - e^{-\lambda t}$
$P_f = 1 - e^{(-87,000 / 95,000)}$
P_f=1-.40
P_f= 0.60 or 60%

783 A major failure requires that two conditions, A and B, exist. Condition A occurs if Event 1 and Event 2 exist. Condition B exists if any one of Events 3, 4, or 5 occur. The probability for Events 1, 2, 3, 4, and 5 are as follows:
E_1 = 0.98
E_2 = 0.95
E_3 = 0.20
E_4 = 0.29
E_5 = 0.50
What is the probability that the major failure will occur?

1) 0.931
2) 0.922
3) 0.056
4) 0.027

Ans.2 To get the answer, E_1 and E_2 are multiplied together to get the probability for Condition A.
P(A) = 0.98 * 0.95
= 0.931
The probability for Condition B to exist is the sum of E_3, E_4, and E_5.
P(B) = 0.20 +0.29 + 0.50 = 0.99
The probability of Condition A and Condition B are then multiplied to get the probability of a major failure.
P(major) = 0.931 * 0.99
= 0.9216

784 The probability of failure of a system is 0.922. The system fault tree requires both "A" and "B" to occur. "A" occurs when Events 1 and 2 occur. "B" occurs when either Events 3, 4 or 5 occur. The probability of each event occurring are as follows:
$E_1 = 0.98$
$E_2 = 0.95$
$E_3 = 0.20$
$E_4 = 0.29$
$E_5 = 0.50$
What is the probability of failure if the events generating "B" had to happen in parallel?

1) 0.922
2) 0.27
3) 0.056
4) 0.027

Ans.4 Condition B events are multiplied together to get the probability of Condition B. Condition A events are also multiplied together to get the probability of Condition A. The respective probabilities of Condition A and Condition B are then multiplied together to get the system failure probability.
P(A) = 0.98 * 0.95 = 0.931
P(B) = 0.20 * 0.29 * 0.50 = 0.029
P(major) = P(A) * P(B)
= 0.931 * 0.029
= 0.027

785 When applying the concept of system safety to fire prevention and protection, the system characteristics include all the following EXCEPT:

1) Boundary
2) Input and output
3) The environment
4) Variables and structure

Ans.3 This does not mean that the system does not interact with the environment, but outside events do not necessarily govern the behavior of the system.

786 In the context of pure systems safety, the fire safety concept tree is similar to:

1) Preliminary hazard analysis
2) Fault tree analysis
3) Failure mode and effect analysis
4) Decision trees

Ans.2 The fire safety concept tree uses "AND" and "OR" gates to determine the probability of events. However, the top event may be a positive action, not a failure or fault.

787 Air brakes on a railroad train is an example of a device that is:

1) Fail-safe
2) Fail-passive
3) Fail-redundant
4) Fail-operational

Ans.1 These brakes are fail-safe and fail-active. If air is lost the brakes are automatically set.

788 A good safety design feature that helps to overcome failures or malfunctions is a:

1) Safety margin
2) Safety factor
3) Maximum foreseeable failure margin
4) Maximum permissible failure factor

Ans.2 By definition, a safety factor is the ratio of the ultimate (breaking) strength of a member, or piece of material, to the actual working stress or to the maximum permissible (safe load) stress when in use. The magnitude of the safety factor depends on how great the cost of failure will be in terms of life or damage.

789 In product safety, the safety professional must give consideration to the design parameters involving malfunctions, operator errors, environmental differences, and:

1) Safety margins
2) Ergonomics
3) Loss minimization
4) Structural failure

Ans.3 The design factors that must be considered include: - minimizing loss - malfunctions - operator error - environmental conditions. MINIMIZING LOSS is a provision of protective methods which can be applied to product safety for the purpose of containment and damage minimization. Examples include bumpers, seat belts, explosion barriers, and other forms of energy absorbing mechanisms or personal protective safety devices.

790 Fail-safe designs try to ensure that a failure will leave the product unaffected or will convert it to a state in which no injury or damage will occur. Which design does NOT meet this criterion?

1) Fail-passive arrangements
2) Fail-active design
3) Fail-operational arrangements
4) Fail-mode design

Ans.4 All of the following designs may be considered fail-safe:
FAIL-PASSIVE ARRANGEMENTS are similar to fuses and circuit breakers; they reduce the system to its first level.. FAIL-ACTIVE DESIGN maintains any condition that will keep the system in a safe mode until corrective or overriding action occurs.
FAIL-OPERATIONAL ARRANGEMENTS allow functions to continue safely until corrective action is possible.

791 A system that is designed to shut down when one or more critical components fail is called:

1) Fail-safe
2) Interlock
3) Circuit breaker
4) Fuse

Ans.1 There are times when a system, guard, elevator, and other high risk equipment is designed to shut down when one or more critical components fail. When this occurs, it will do so in a safe position or condition without the experience of a major loss.

792 The interaction of component parts to perform a function or task that accomplishes something defines:

1) An analysis
2) A system
3) A planning method
4) A subsystem

Ans.2 A system is an orderly arrangement of components that are interrelated and that act and interact to perform some task or function. As parts of a system, the components usually interact and complement each other. However, it is essential to recognize that a malfunction of one component can affect others, which impacts the total system.

793 A key advantage of Boolean algebra is considered to be:

1) It is superfluous
2) It is deceptively simple
3) It can utilize AND/OR/NOT connectors
4) A calculator is not needed

Ans.3 Boolean algebra is widely used to develop probabilities involving AND/OR/NOT situations.

794 An example of a fail-passive device is the:

1) Fuse
2) Emergency generator
3) High temperature limit switch
4) Both 1 and 3

Ans.1 A fuse would be the best example of a fail passive device as its failure brings the system to its lowest energy state (no current flow). The other devices do not necessarily bring the system to its lowest energy state given a failure.

795 This system safety design philosophy, given a failure, keeps the system energized in a safe mode until corrective or overriding action can take place:

1) Intrinsically safe
2) Fail-passive
3) Fail-active
4) Fail-operational

Ans.3 The fail-active design philosophy keeps the system energized and in a safe mode until corrective action or an override can take place. Intrinsic safety eliminates failure, fail passive knocks the system to its lowest energy level, and fail operational allows the system to operate without interruption.

796 The following system is an example of a fail-active design:

1) Cylinder with rupture disc
2) Class I, Division 1 wiring
3) Chirping smoke detector
4) Either 2 or 3

Ans.3 A fail-active design normally creates redundancy by the use of standby equipment. A chirping smoke detector is the classic example of a fail-active design. When the battery starts to run down, most ionizing smoke detectors for home use will chirp periodically to warn the occupants that the battery is failing. The smoke detector may not function (depending upon the actual battery status), but the chirp design actively informs the occupant that a failure has occurred or will soon occur.

797 A system safety design concept that allows functions to continue safely given a failure is:

1) Intrinsically safe
2) Fail-active
3) Fail-passive
4) Fail-operational

Ans.4 A fail-operational design allows a system that is experiencing a failure to continue to operate safely until corrective action is possible. Limited disruption, if any, is experienced.

798 Water feed valves for boilers that remain open during a control failure to allow the boiler to operate and not build up excess pressure are:

1) Intrinsically safe devices
2) Fail-passive devices
3) Fail-active devices
4) Fail-operational devices

Ans.4 Water feed valves that remain open during boiler control failure, allowing the boiler to operate and not build up excess pressure are fail-operational devices.

799 An example of a non-fail-safe device would be:

1) Any single point failure
2) Deadman switches
3) Circuit breaker
4) GFCI

Ans.1 Any single point failure is an example of a non-fail-safe device.

800 A battery operated smoke detector is an example of what type of design?

1) Interlock
2) Fail-active
3) Fail-passive
4) Fail-operational

Ans.2 A battery operated smoke detector is an example of a fail-active design.

801 An "AND" gate implies:

1) Boolean division
2) Boolean addition
3) Boolean multiplication
4) Boolean introspection

Ans.3 If A and B are required for C to operate, and A and B are joined by an "AND" gate, then the probability of C is AB.

802 How would you perform the Boolean connection of "heat", "fuel", and "oxygen" to the output "fire?"

1) "AND"
2) "OR"
3) "NOR"
4) "NOT"

Ans.1 As you must have "heat" AND "fuel" AND "oxygen" to have a fire, the "AND" gate is necessary.

803 The technique of probabilistic risk assessment was developed by what industry?

1) Electronics
2) Transportation
3) Aerospace
4) Nuclear power

Ans.4 Many of the techniques in PRA were applied and developed by the nuclear power industry in the 1970's. The techniques were developed to estimate failure rates, develop maintenance schedules, assess population safety, etc.

804 Which of the following is NOT correct Boolean logic?

1) AA = A
2) A + A = A
3) A + 1 = 1
4) A + (AB) = AB

Ans.4 A + (AB) = A (A + B)

805 A current problem with risk and its perception is that the public's estimation of risk _____ the government or experts.

1) Is often not the same as
2) Is much less valid
3) Is usually not financially acceptable
4) Cannot be calculated by

Ans.1 The public's perception of risk associated with a given technology may vary widely from the perception of the government or risk experts. The view may be based on partial information. The differences are real and must be addressed as thoroughly and carefully as the scientific analyses associated with the risk assessment.

806 The least likely term which describes designing the job to fit the worker is:

1) Human factors
2) Human engineering
3) Ergonomics
4) Human systems
5) \

Ans.4 Ergonomics, human factors, human engineering, biomechanics, and engineering anthropometry are terms common to the objective leading to designing a work station, display, or job, to fit the worker and avoid unnecessary environmental stresses. The desired results of this design is a decrease of injury/illness and an increase in productivity.

807 What is the most appropriate definition of human factors engineering?

1) Discipline dealing with man/machine interface
2) Discipline related to industrial psychology
3) Discipline seeking to improve worker happiness
4) Discipline to increase worker efficiency

Ans.1 Human factors engineering is the discipline which is concerned with the man/machine interface. Factors it considers include sensory inputs, operator response, operator movements, work/rest cycles, and whether speed and accuracy are required.

808 As part of a valid and complete hazard communication program, the employer must do all of the following EXCEPT:

1) Prepare MSDS's for all substances in the process
2) Have MSDS's for substances with employee exposure
3) Have a written Hazard Communication Program
4) Train employees in the hazards of the chemicals

Ans.1 Employers are not required to prepare MSDS's for the substances they use. The manufacturer has that responsibility. The employer is required to have MSDS's on hand and readily available to the employee for all the substances to which he or she is exposed.

809 One characteristic of system safety is that it includes all other types of safety. A second characteristic of system safety is that:

1) It is performed before mishaps occur
2) It is effective only on complex designs
3) It is most effective during testing
4) There is no second characteristic of system safety

Ans.1 System safety is always conducted before mishaps actually occur. This factor, plus the fact that the system safety practitioner may use information from all other safety areas, makes this discipline unique.

810 Systems safety analytical techniques to determine foreseeability include:

1) Use of biorhythms
2) Use of fault finding
3) Use of statistical techniques including determination of mean, mode, standard deviation
4) Use of preliminary, failure modes and effects, fault hazard, and fault-tree analysis

Ans.4 All the techniques in Answer 4 are legitimate techniques for the safety professional. Biorhythms were used in Japan with questionable results for predicting bus accidents. Fault finding is not a legitimate technique. Statistical techniques may be used as part of the analyses done when applying other appropriate systems safety techniques.

811 Risk is an expression of probable loss over a specific time period. It can be expressed in accidents, deaths, dollars, or other units. If a company operates a fleet of 25 trucks worth $22,000 each, and each operates 1,600 hours per year, what is the risk due to total loss per year if the replacement cost for a new truck is 1.2 times the old truck, and accidents occur 4.5 times per 10,000 hours of operation?

1) $118,800 per year
2) $475,200 per year
3) $396,000 per year
4) $2,970,000 per year

Ans.2 Risk = D * M * N where
D = loss per accident
M = Accident rate
N = time of measurement

Risk = (1.2 * $22,000 / truck)(4.5 x 10^{-4} per hour) (25 trucks * 1,600 hr/yr)
= $475,200 / yr

812 System safety is defined as:

1) A method to detect and eliminate potential hazards
2) An engineering method
3) An analysis of major components
4) A statistical analysis of the safety system

Ans.1 Systems safety is the utilization of systematic scientific engineering and management principles to identify and minimize potential hazards.

813 Within a Management Oversight Risk Tree (MORT) analysis, what is NOT considered part of the management system factors?

1) Policy
2) Implementation
3) Risk assessment
4) Barriers LTA

Ans.4 Barriers are considered part of specific control factors.

814 Within a Management Oversight Risk Tree (MORT) analysis, what is NOT considered part of specific control factors?

1) Amelioration LTA
2) Persons or objects in an energy path
3) Assumed risks
4) Unwanted energy flow

Ans.3 Assumed risks are considered as a separate tree path.

815 Which is not part of the risk assessment system defined within a MORT analysis?

1) Safety program review
2) Hazard analysis process
3) Unwanted energy flow
4) Information systems

Ans.3 Unwanted energy flow is considered under specific control factors and is not a part of the risk assessment system defined within a MORT analysis.

816 TRADITIONAL safety program components utilized to influence employee attitudes and behavior are identified with:

1) Written procedures
2) Posters, booklets, awards
3) Education and training
4) Provision of free personal protective equipment

Ans.2 Traditional safety programs attempt to communicate with employees for the purpose of motivating them to change their attitudes, or better yet, their behavior. To do this, posters, booklets, slide shows, displays, signs, bulletin boards, awards, dinners, and prizes are utilized.

817 You have been appointed safety director of a company with 4 plants. These plants are subject to annual safety audits which are scored. You want to determine if success in the audit program correlates to success in injury prevention. Based on the data provided in the Table below, what is the Spearman rank coefficient of correlation between the safety audit score and injury rates?

	PLANT A	PLANT B	PLANT C	PLANT D
AUDIT SCORE	75	86	45	90
INJURY RATE	3.2	4.4	4.8	5.6

1) 1.4
2) 0.40
3) -0.40
4) -0.60

Ans.3 First, determine the difference in rank of each set, and square that difference. Use a chart like this one.

Audit Score Injury Rate				
Plant	Rank	Rank	D	D^2
A	3	1	2	4
B	2	2	0	0
C	4	3	1	1
D	1	4	-3	9

The sum of the squares of the differences is
$\Sigma D^2 = 4 + 0 + 1 + 9 = 14$

The symbol "N" represents the number of data points. N is 4.

Use the following formula to calculate the Spearman rank:

$\rangle = 1 - 6 \Sigma \{D^2 / [N (N^2 - 1)]\}$
$= 1 - 6 \{14 / [4 (4^2 - 1)]\}$
$= 1 - 1.40$
$= -0.40$

818 If a Spearman rank coefficient = -0.4 for injury rates and plant audit scores, what is your conclusion regarding the correlation between the injury rates and audit scores?

1) There is a weak direct correlation
2) There is a strong direct correlation
3) There is no meaningful correlation
4) It is not possible to correlate injury rates and audit scores

Ans.3 The Spearman rank is a reflection of correlation between two sets of numbers. Since the Spearman rank is negative, the injury rates and audit scores are mathematically inversely related. When dealing with plant scores and injury rates, though, this inverse relationship is somewhat meaningless. If taken with mathematical literalness, the Spearman rank of -0.4 implies that there is a weakly related INCREASE in injury rates when the plant audit scores are good.

819 A study collected 161 death certificates of former workers at a plant. Of these 161 deaths, 38 workers died of cancer where 27.9 would normally be expected to do so. The rest of the workers died of other causes where where 133.1 would normally be expected to do so. Compute the $?^2$ (chi-square) statistic.

1) 2.79
2) 4.42
3) 9.21
4) 13.3

Ans.2 $?^2 = \Sigma [(\text{observed-expected})^2 / \text{expected}]$
$= (38 - 27.9)^2 / 27.9 + (123 - 133.1)^2 / 133.1$
$= 4.42$

Explanation of the $?^2$ Statistic:

We have only one degree of freedom in this study since we can only set one of the two death rates (the second rate is dependent on the first). Use the Upper Percentage Points of the $?^2$ Distribution table and determine the "p" associated with a degree of freedom of 1 and a χ^2 statistic of 4.42.

The "p" will be between 0.05 and 0.01 (closer to 0.05). If you want a better estimate of "p," you could interpolate to derive a better estimate based on this $?^2$ statistic.

The "p" of approximately 0.05 means that the data are statistically significant at the p = 0.05 level. In other words, there is only a probability of 0.05 that the data are NOT significant.

820 In behavioral terms, a safety poster:

1) Is an activator
2) Is "visual noise"
3) Has no impact on behavior
4) All of the above

Ans.1 As an activator which serves to direct desired behavior, safety posters have only a small impact on behavior.

E. Scott Geller, The Psychology of Safety